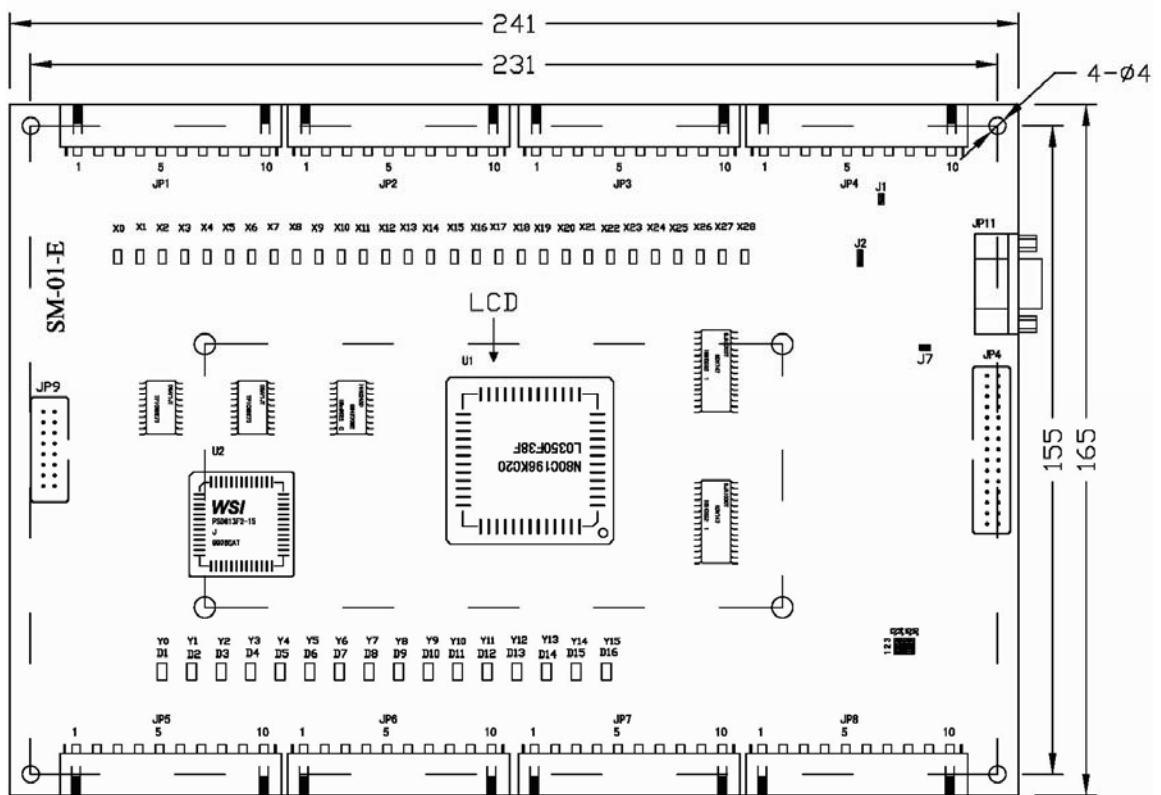


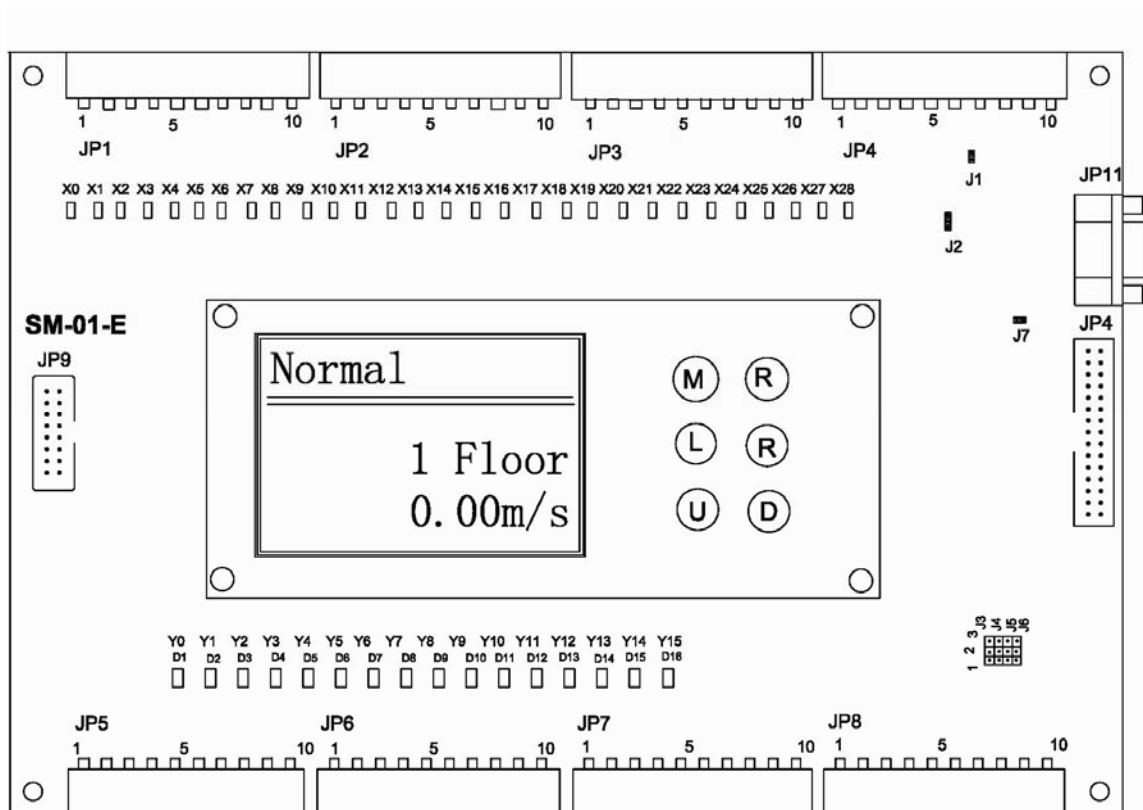
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1. Physical appearance and installation dimensions



2. Ports



2.1. Brief introduction of input LED

□ □ □ □ □ □ □ □ □ □ □ □ □ □ ... □ □ □
 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 ... 27 28 29

The indicating lights at upper row, from left to right are 1, 2, 3……28, 29;
 Corresponding inputs are X0, X1, X2……X24, X25, X26, X27, X28.

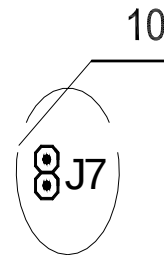
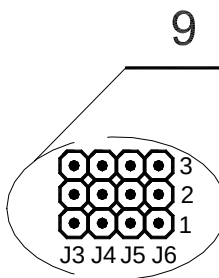
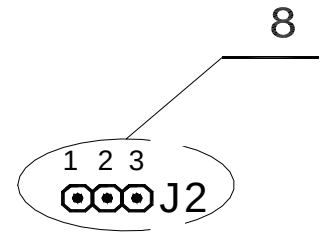
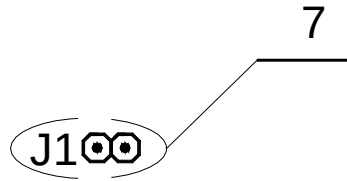
2.2. Brief introduction to output LED

□ □ □ □ □ □ □ □ □ □ □ □ □ □ □ □
 D1 D2 D3 D4 D5 D6 D7 D8 D9 D10 D11 D12 D13 D14 D15 D16

The indicating lights at lower row, from left to right are D1, D2, D3……D15, D16.
 Corresponding outputs are Y0, Y1, Y2……Y14, Y15.

2.3. Description of jumper

It is always neglected the importance of configuring the system by using jumper settings. Improper setup may cause many problems. The physical appearance of the jumper and their designation are described individually as below.



Jumper	Designation	Remark
J1	To connect terminal resistance for Duplex, group control, J1 is always connected during the elevator is duplex, group controlled, which connect the built-in communication terminal resistor, for strengthening the system's anti-interference ability	
J2	J2 is used for the selection of RS232/RS485. Make a jumper connection <J2.1-J2.2>, communication mode is RS232, MODEM and telephone wire could be equipped with, for proceeding the remote monitoring. Make the jumper connection <J2.2-J2.3>; communication mode is RS485, for proceeding the concentrated monitoring. It is necessary to connect J7 when selecting, connect it with the terminal resistance in order to strengthen the system's anti-interference ability.	
J3、J4 J5、J6	JP7 connects when using the differential encoder, suitable to CT, KEB etc. speed regulators. At this time, connect jumpers <J3.1-J3.2>, <J4.1-J4.2>, <J5.1-J5.2>, <J6.1-J6.2>. When using collector open circuit output or push-pull output encoders, JP8 is connected, please make the jumper connections <J3.1-J3.2>, <J4.1-J4.2>, <J5.2-J5.3>, <J6.2-J6.3>.	
J7	RS485 communication terminal resistance jumper, for anti-interference purpose, at this time J2 should jump to connect <J2.2-J2.3>.	

3. Electrical Specification

- ◆ Output of switching power supply

Rated value	DC24V	DC5V
Permissible range	DC24V $\pm$ 0.3V	DC5V $\pm$ 0.1V
Input current	1.8A(<25 th floor)	2A

If a building is less than 25-storey, the switching power supply could be configured according to the above table. If a building is more than 25-storey, an extra switching power supply should be added. For a 40-storey building, the input current of DC24V is 4A.

- ◆ Digital Input

Input point amount		29 (removable terminal)	
Input form		Optical coupling	
Input voltage	Rated value	24VDC	
	"1" signal	X0-X25	12~24VDC
		X26-X28	AC110V/DC80V
	"1" signal	0~5VDC	
Input current	"0" signal	0~2mA	
	"1" signal	4~7mA	
Isolating group (common end)		1	
Time delay		10ms	
Input frequency limit		1KHz	
Cable length	Screened	400m	
	Non-screened	200m	

- ◆ Digital Output

Output point amount		16 (removable terminal)
Form of connecting point		Relay output
Load voltage	Rating	24VDC/24~230VAC
	Permissible range	5~30VDC/20~250VAC
Contact switch capacity	Resistive load	5A
	Inductive load	3A
	Light load	100W
Running life	Mechanical life	
	Under rated load	
Cable length	Non-screened	200 m
	Screened	400 m
Insulation		24VDC~230VAC

♦ Analogue output

Point amount			2（removable terminal）
Output range	V1	Rated value	0~+10VDC
		Range	Maximum +9.5~10VDC
	V2	Rated value	-10~+10VDC
		Range	Minimum -10~-9.5VDC, Max.+9.5~10VDC
Load impedance			>10K Ω
Resolution			8 digit
Basic error			±5%

♦ Communication port (CAN1 and CAN2)

Connecting point form	Removable terminal
Signal form	Differential voltage
Communication manner	CAN Bus
Max. Time delay of communication	10ms

♦ High speed processing port (JP7 and JP8)

Connecting point form	Removable terminal
Signal form	Applicable to collector open circuit or push-pull output signal or differential signal encoder.
Max. Baud rate	< 100K

♦ RS232/485 communication port

Connecting point form	9-way D socket
Communication method	Standard RS232 or standard RS485

4. Definitions of Input/Output Interface

4.1. Definition of external input interface

JP1, JP2 & JP3 terminals are input interfaces of external switching signals, in which JP3.9 and JP3.10 requires the external +24V power supply input, it is used as the isolating power supply for the external input signal. The following table outlines the particular definitions of their input point individually.

Socket No.	Ter. No.	I/O type	Definition	Input type	LED
JP1	JP1.1	Input	X0, inspection signal, Open is inspection, CLOSE is automatic	Default	1
	JP1.2	Input	X1, Up driving direction signal, when in inspecting CLOSE means inching up, when in attendant mode, it means change driving direction to up driving	Default	2
	JP1.3	Input	X2, Down driving direction signal, when in inspecting CLOSE means inching down, when in attendant mode, it means change driving direction to down driving	Default	3
	JP1.4	Input	X3, up multi-floor terminal slow down switch, elevator of 2m/s needs to use it.	N/C	4
	JP1.5	Input	X4, down multi-floor terminal slow down switch, elevator of 2m/s needs to use it.	N/C	5
	JP1.6	Input	X5, up limit switch	N/C	6
	JP1.7	Input	X6, down limit switch	N/C	7
	JP1.8	Input	X7, up single floor terminal slow down switch	N/C	8
	JP1.9	Input	X8, down single floor terminal slow down switch	N/C	9
	JP1.10	Input	X9, up leveling switch	N/O	10
JP2	JP2.1	Input	X10, Down leveling switch	N/O	11
	JP2.2	Input	X11, failure output signal of speed regulator	N/O	12
	JP2.3	Input	X12, Fire fighting switch	N/O	13
	JP2.4	Input	X13, Safety loop relay detection	N/O	14
	JP2.5	Input	X14, Door lock loop relay detection	N/O	15
	JP2.6	Input	X15, Speed regulator's in-line contactor detection	N/O	16
	JP2.7	Input	X16, Speed regulator's out-line contactor detection	N/O	17
	JP2.8	Input	X17, Braking contactor detection	N/O	18
	JP2.9	Input	X18, Door zone signal input, used for releveling and advance leveling with opened door	N/O	19
	JP2.10	Input	X19, Speed regulator's running signal detection, when detected this signal, the brake will be released	N/O	20

JP3	JP3.1	Input	Input X20, spare	N/O	21
	JP3.2	Input	Input X21, spare	N/O	22
	JP3.3	Input	Input X22, spare	N/O	23
	JP3.4	Input	Input X23, spare	N/O	24
	JP3.5	Input	Input X24, spare	N/O	25
	JP3.6	Input	Input X25, spare	N/O	26
	JP3.7		X0-X25 common terminal of input signal		
	JP3.8		X0-X25 common terminal of input signal		
	JP3.9		X0-X25 negative end of isolated circuit power supply, 0V		
	JP3.10		X0-X25 positive end of isolated circuit power supply, 24V		

Comparison table of the input ports corresponding to its **LED** display

Input	X0	X1	X2	X3	X4	X5	X6	X7	X8	X9
LED	1	2	3	4	5	6	7	8	9	10

Input	X10	X11	X12	X13	X14	X15	X16	X17	X18	X19
LED	11	12	13	14	15	16	17	18	19	20

Input	X20	X21	X22	X23	X24	X25	X26	X27	X28	
LED	21	22	23	24	25	26	27	28	29	

4.2. Definition of main board's power supply

JP4 terminal r is the interface supplying power supply to the main controller. Its power is supplied by the external switching power. The following table outlines their definitions:

Socket No.	Terminal No.	Definition
JP4	JP4.1	0V power supply of main controller
	JP4.2	+24V power supply of main controller
	JP4.3	0V power supply of main controller
	JP4.4	+5V power supply of main controller

4.3. Definition of main board's communication interface

JP4 terminals are located on the upper right corner of the main controller's front view. JP4 is a communication terminal connecting to the hall calls board and car board and must be used in twisted pair. TXV1+, TXV1- one strand of twisted pair is used, and TXA1+, TXA1- use another strand of twisted pair. JP4 is also used as the duplex, group

control interface connecting to JP4.6, P4.7 and JP4.8 three wires of the two main controllers correspondingly when used for duplex control, in which TXA2+, TXA2- must use one strand of twisted pair too. The following table outlines their definitions:

Socket	Terminal	Definition
JP4	JP4.2	Positive end of hall call and car serial communication power supply, TXV1+
	JP4.1	Negative end of hall call & car serial communication power supply, TXV1-
	JP4.9	Positive end of hall call and car serial communication signal, TXA1+
	JP4.10	Negative end of hall call and car serial communication signal, TXA1-
	JP4.5	Positive end of duplex or group control serial communication power supply, TXV2+
	JP4.6	Positive end of duplex or group control serial communication signal, TXA2+
	JP4.7	Negative end of duplex or group control serial communication signal, TXA2-
	JP4.8	Negative end of duplex or group control serial communication power supply, TXV2-

4.4. Definition of safety detection interface

JP5 terminal is located at the lower corner of the main controller for the detection of safety loop and door lock loop; it is an input terminal of +110V. The following table outlines their definitions:

Socket	Terminal	I/O type	Definition
JP5	JP5.1	Input	X26, positive end of safety loop detection, 110V input
	JP5.2	Input	0V end of input X26
	JP5.3	Input	X27, positive end of door lock loop detection, 110V input
	JP5.4	Input	0V end of input X27
	JP5.5	Input	X28, spare, positive end, 110V input
	JP5.6	Input	0V end of input X28

For input Xn: If the input point of Xn is normally on signal, the LED corresponding to Xn is lit. If this input point Xn is a normally off signal, the corresponding LED will be off.

4.5. Definition of main board's output interface

Some ports of JP5, JP6, JP7 are the main board's output interfaces. The state of the two terminals was the transition form switched by the relay. Definitions of each point is shown in the following table:

Socket No.	Ter. No.	I/O type	Definition	LED
JP5	JP5.8	Output	Output relay Y0, Brake contactor output	D1
	JP5.9	Output	Output relay Y1, Brake forced excited contactor output	D2
	JP5.10	Output	Output relay Y2, speed regulator in-line contactor output	D3
JP6	JP6.1	Output	Output relay Y3 speed regulator out-line contactor output	D4
	JP6.2		Output relay Y0, Y1, Y2, Y3 common end	
	JP6.3	Output	Output relay Y4, door opening relay output	D5
	JP6.4	Output	Output relay Y5, door closing relay output	D6
	JP6.5	Output	Output relay Y6, door open in advance or door open then leveling relay output	D7
	JP6.6	Output	Output relay Y7, spare	D8
	JP6.7		Output relay Y4, Y5, Y6, Y7 common end	
	JP6.8	Output	Output relay Y8, spare (door close relay1)	D9
	JP6.9	Output	Output relay Y9, spare (door open relay1)	D10
	JP6.10		Output relay Y8, Y9 common end	D11
JP7	JP7.1	Output	Output Y10, speed regulator up direction	D12
	JP7.2	Output	Output Y11, speed regulator down direction	D13
	JP7.3	Output	Output Y12, speed regulator running function	D14
	JP7.4	Output	Output Y13, speed regulator multi-step speed terminal 1	D15
	JP7.5	Output	Output Y14, speed regulator multi-step speed terminal 2	D16
	JP7.6	Output	Output Y15, speed regulator multi-step speed terminal 3	
	JP7.7		Common end of Output terminal Y10-Y15	

For output Y_n: If the normally illuminated LED of Y_n is lit, it shows that the relay corresponding to Y_n output point is actuated, i.e. Y_n outputs is sending an "ON" signal. If Y_n is connected to a contactor or relay, the coil of contactor or relay is energized.

4.6. Given analog speed and torque compensation

Some ports of JP7 are the output of given analog speed signal and the output of analog load compensation signal. The following table explained their definition:

Socket No.	Ter. No.	I/O type	Definition
JP7	JP7.8	Output	Analog load compensation signal, output to torque compensation end of the speed regulator, $\pm 10V$ signal
	JP7.9	Output	Given analog speed, output to speed setting end of the speed regulator, 0-10V signal
	JP7.10		Analog signal 0V

4.7. Definition of the encoder's input interface

JP8 that are at the lower right corner of the main controller, are the input interfaces of the encoder. The following table outlines their definitions:

Socket No.	Terminal No.	Definition
JP8	JP8.1	Encoder power supply +5V or+ 15V
	JP8.2	Encoder power supply 0V
	JP8.5	Encoder A phase, could accept collector open circuit output or push-pull output, acceptable frequency is 0-100 KHz
	JP8.6	Encoder B phase, could accept collector open circuit output or push-pull output, acceptable frequency is 0-100 KHz
	JP8.7	Differential encoder A+
	JP8.8	Differential encoder A-
	JP8.9	Differential encoder B+
	JP8.10	Differential encoder B-

4.8. Description of other interfaces' definitions

JP10: LCD Humanized man-machine interface. Its LCD operation panel not only could display the elevator's speed, direction, status, but also display the elevators given speed curve and feed back speed curve. In addition, elevator's parameters could be set and elevator's failure records could be inquired when needed.

JP11: RS232/RS485 MODEM remote monitoring interface. The remote control center could directly monitor the situations of site elevators. Thus, it greatly improves the service efficiency and quality of after sales. Their definitions for each pin are detailed in the following table:

Pin	Definition	Pin	Definition
JP11.1	DCD	JP11.6	RS485-A
JP11.2	RXD	JP11.7	RS485-B
JP11.3	TXD	JP11.8	X
JP11.4	DTR	JP11.9	X
JP11.5	SGND		

5. Connecting method

5.1. Power supply connecting

The working power supply of the main controller was stabilized by a reliable switching power supply. Two groups of DC 24V, and DC 5V outputs are provided.

Item	Requirements or remarks
Application	JP4 is the main controller's working power supply provided by the main controller's power supply terminal
Connector form	CH2.54-6
Interface definition	Pin 1、3 are COM, pin 4 is +5V, pin 2、5 is +24V, pin 8 is COM

Caution: The switching power supply must be connected to a reliable ground; otherwise it may cause the main controller work abnormally.

5.2. Car and landing communication connection CAN1 (JP4)

JP4 is labeled in the upper right corner of the main controller. It is the communication interface of the main controller with the car controller and Indicator/call board.

Item	Requirements or remarks
Application	CAN1, JP4 used as the communication interface of the main controller with the car controller and indicator/call board
Connector form	4 ways male connector on the main controller, matched with 4 ways female connector
Interface definition	Pin 10 is TXA1-, pin 9 TXA1+, pin 8 is TXV1-, pin 5 is TXV1+
Requirements of connecting wire	Must be connected with twisted pair, TXA1+, TXA1- is one set twisted pair, TXV1+, TXV1- is one set twisted pair. Stranding pitch is $\leq 30\text{mm}$.
Particular	As shown on the following diagram

5.3. Duplex or group control communication connection

JP4 is labeled on the main controller. It is used as a mutual communication interface when under the duplex control of two elevators, or group control of multi-elevators.

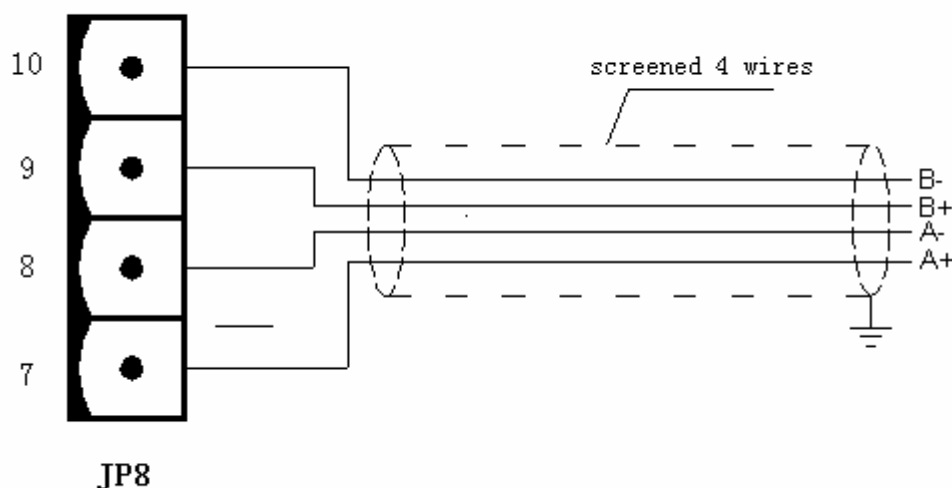
Item	Requirements or remarks
Application	CAN2 (JP5) is used as the mutual communication interface during the duplex control of two elevators, or the group control of multi-elevators.
Connector form	4 ways male connector on main controller, matched with 4 ways female connector
Interface definition	Pin 7 is TXA2-, pin 6 is TXA2+, pin 8 is TXV-
Requirements of connecting wire	Must be connect with twisted pair, TXA+, TXA- is one set twisted pair, stranding pitch is $\leq 30\text{mm}$.

5.4. High speed processing interface of ENCODER

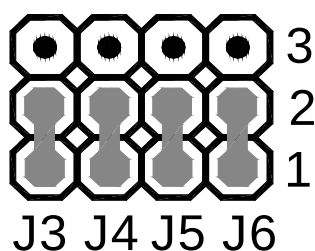
JP8 lies on the main controller. It receives a feed back signal from the rotary encoder for evaluating the elevator's position.

Item	Requirements or remarks
Application	JP8 receives the feed back signal from the rotary encoder for evaluating the car's position
Connector form	4 ways male connector on main controller, matched with 4 ways female connector
Interface definition	For the concrete definition, see the description of above section.
Requirements of connecting wire	Must be connected with screened wire.
Particular wiring method	As shown on the following diagram

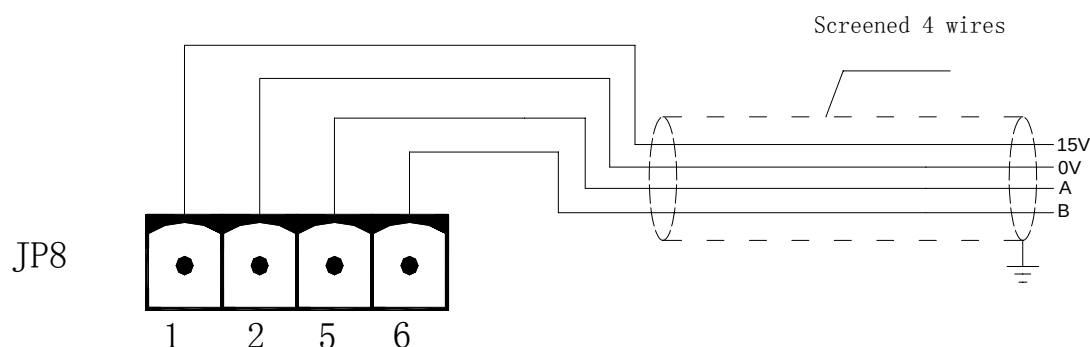
I. When the encoder is a differential encoder, the wiring is shown as follows:



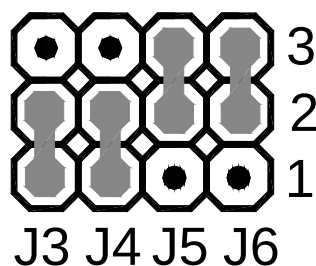
Setup of jumping is as follows:



II. When the encoder is a collector open circuit or push-pull output type encoder, the wiring is shown as the following diagram:



The jumper is setup as following:



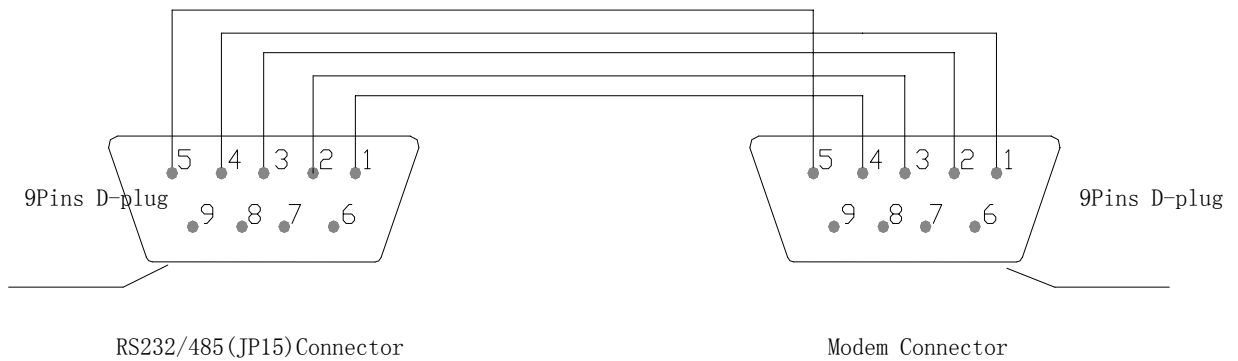
5.5. RS232/485 connector for remote monitoring.

Item	Requirements or remarks
Application	RS232/485, JP11 used for connecting the remote monitor via MODEM, also used as the notebook computer's commissioning interface during commissioning
Connector form	D type 9 ways male connector on main controller, matched with D type 9 ways female connector
Interface definition	See table bellows
Requirements of connecting wire	
Particular wiring method	As shown on the following diagram

● Connecting the wires between RS232/485 and MODEM during monitoring

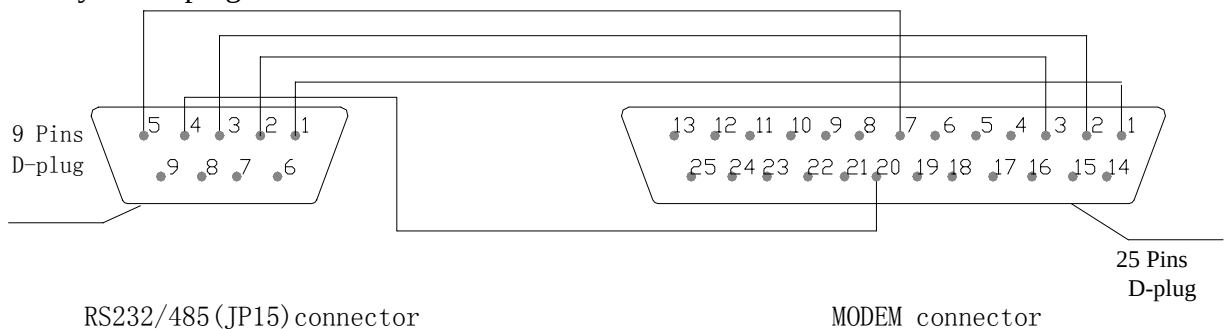
A. RS232 connection

I. The interface terminal of **SM-Sup** is a 9 ways plug; the interface of MODEM is a 9-ways female plug



SM-Sup	MODEM	Remark
1	1	DCD
2	2	RXD
3	3	TXD
4	4	DTR
5	5	SGND

II. The interface of SM-Sup is 9 ways female plug; the interface of MODEM is 25 ways male plug



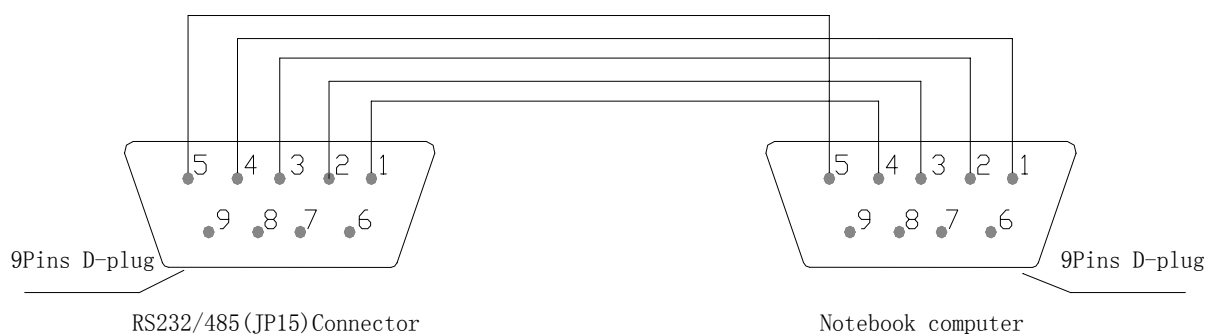
RS232/485 (JP15) connector

MODEM connector

SM-Sup	MODEM	Remarks
1	1	DCD
2	3	RXD
3	2	TXD
4	20	DTR
5	7	SGND

B. RS485 connection See attachment 2

● Wire connection for notebook computer commissioning



RS232/485 (JP15) Connector

Notebook computer

SM-Sup	Computer	Remarks
1	4	DCD
2	3	RXD
3	2	TXD
4	1	DTR
5	5	SGND

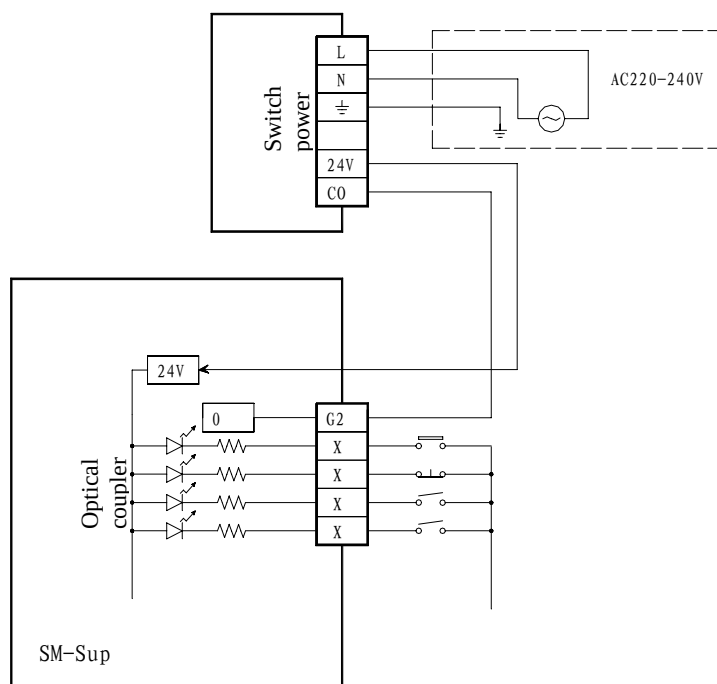
5.6. Scheme of Input/Output external wire distribution

The following is a scheme of **SM-Sup** main controller's internal signal circuit and external wire distribution method.

Note: Terminal number is according to actual schematic drawing.

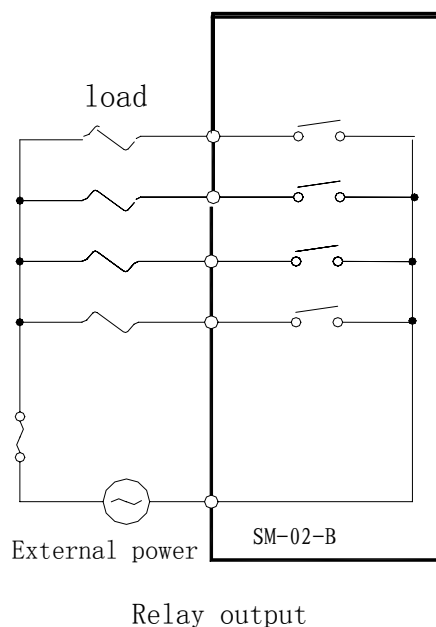
● Scheme of input external wire distribution

The basic wiring method of the input loop is shown in following diagram.



● Scheme of output external wire distribution

The basic wiring method of the output loop is shown in following diagram.



● Analog Signal Giving

The JP7.8, JP7.9 that labeled on the Main Controller, its common terminal is JP7.10

Item	Requirements or remarks
Application	V1 (JP7.9) is used for the output of elevator speed command. It is connected directly to the analog input of the encoder, for controlling the elevator's running speed. The voltage range of V1 (JP7.10) relative to 0V (JP7.9) is DC 0V~10V. 0V shows the speed command is 0. 10V—the elevator's speed is the rated speed. V2 (JP7.8) is used for the analog load compensation signal output.
Connector form	Plug-in terminal
Interface definition	
Requirements of connecting wire	Must use 2-core screened wire, passing through the magnetic ring near the encoder; the screened layer connects to 0V terminal.
Particular wiring method	As shown on the following diagram below

Below is the wiring scheme of the analog given speed with other inverter. The magnetic ring near the inverter is excluded from the scheme, and should be added when operating.

Note: Terminal number is according to actual schematic drawing.

A) Wiring scheme of analog given speed with (YASKAWA) inverter



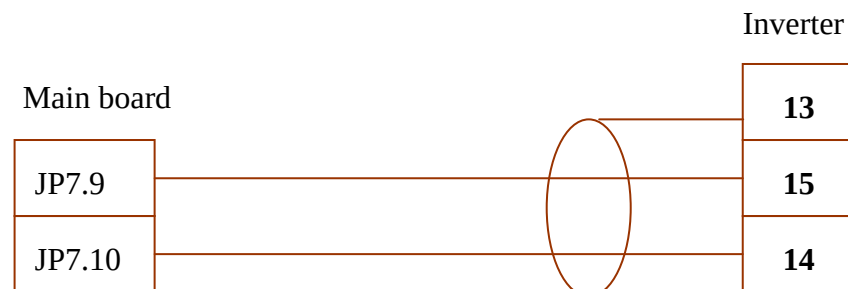
B) Wiring scheme of analog given speed with (FUJIG11UD) Inverter



C) Wiring scheme of analog given speed with (UNIDRIVE) Inverter



D) Wiring scheme of analog given speed with (KEB) Inverter



● Wiring scheme of digital speed output

The arrangement of the speed regulator's digital multi-stage speed terminal is as follows:

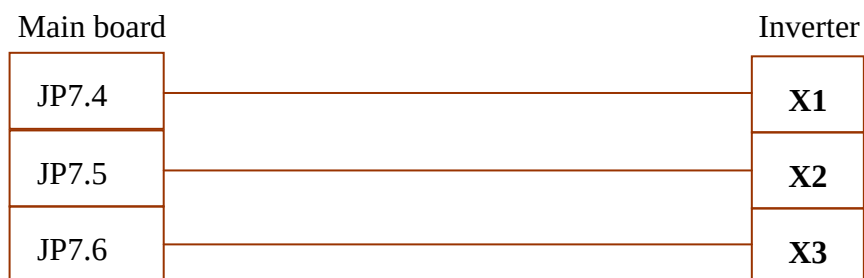
Y13	Y14	Y15	expressing speed
0	0	0	Stop
1	1	0	Climb
0	0	1	Inspection
1	0	1	Single floor
0	1	1	Double floor
1	1	1	Multi-floor

The wiring scheme of given digital speed with the inverter is given below.

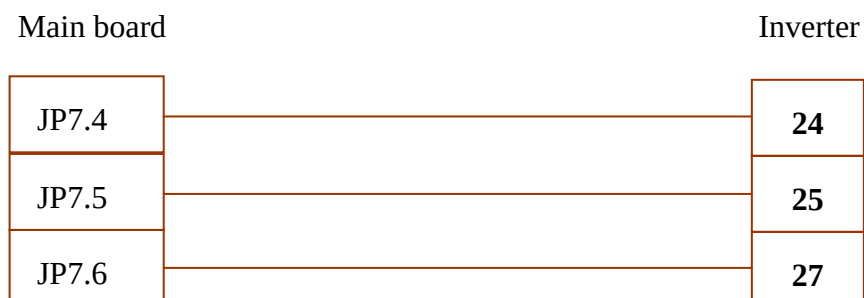
A) Wiring scheme of a given digital speed Inverter for (YASKAWA)



B) Wiring scheme of given digital speed Inverter for (FujiG11UD)



C) Wiring scheme of given digital speed Inverter for (UNIDRIVE)



5.8. Application—connecting with the inverter

The control of the main controller of the speed regulator (Frequency Inverter) consists of four parts. They are the main loop, control loop, analog given speed loop (voltage signal) or given digital speed (digital signal) and encoder feedback loop respectively.

The main loop means the 3-phase AC 380V loop, including AC380V power supply inlet, main contactor, auxiliary contactor, speed regulator and motor.

The control loop controls the contactor's ON/Off of the main loop, for controlling the put into and cut off of the speed regulator with the motor.

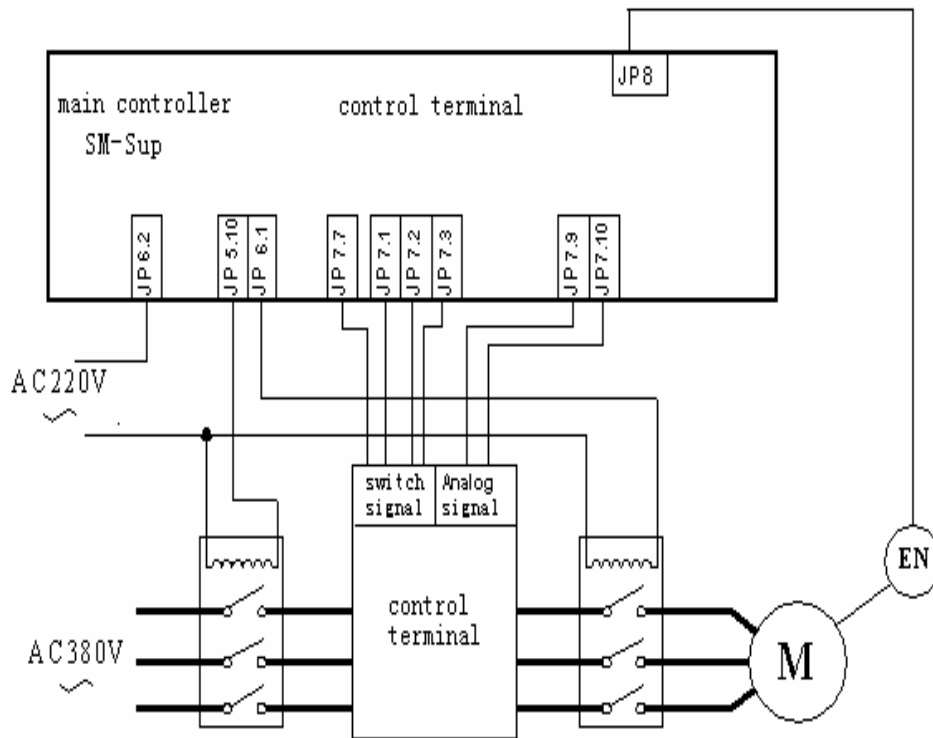
The main controller is connected to the speed regulator's control terminal, for giving out running commands of the running, up/down direction, and making the speed regulator execute the stop, forward running, reverse running actions according to commands.

The main controller's analog output terminal (JP7.9 and JP7.10) is connected to the speed regulator's analog input control terminal which controls the motor's rotation speed according to the analog given voltage. The main controller's digital given speed (JP7.4, JP7.5, JP7.6) is connected to the speed regulator's digital input terminal, which controls the motor's running according to the digital code assignment.

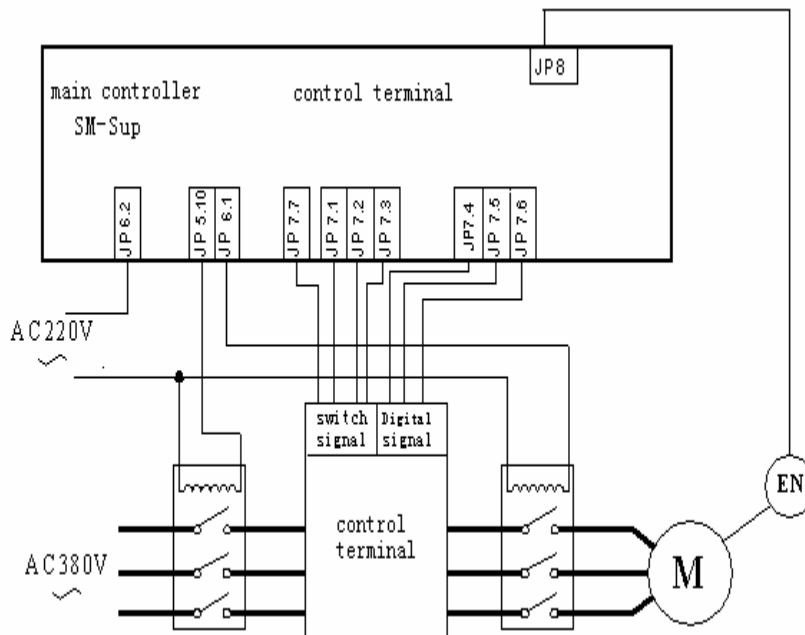
The main controller's high speed processing interface (CODER, JP8) accepts the rotary encoder's feedback to give the real time running curve.

In addition, the main controller gives the running signal of the inverter, detects the fault signal of the inverter and takes the remedy action.

A) During the state of given analog speed



B) During the state of given digital speed:



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